# ISC Class 11 Computer Science – Term 1 Model Question Paper 2

Subject: Computer Science
Paper: 1 (Theory)
Maximum Marks: 60
Time Allowed: 1 Hour 15 min
(Candidates are allowed 2 minutes for reading the paper. They must NOT start writing during this time.)

## Instructions:

1. This paper is divided into two parts: Part I and Part II.
2. Answer all questions in Part I.
3. Answer two questions from Section A and two questions from Section B of Part II.
4. All working, including rough work, should be done on the same sheet as the rest of the answer.

## PART I (20 Marks)

Answer all questions in this part.

### Question 1 [10 × 1 = 10]

(i) The complement of the expression A + B·C according to De Morgan's laws is:
(a) A′·(B′+C′)   (b) A′+(B′·C′)   (c) (A′+B′)·(A′+C′)   (d) A′·B′+C′

(ii) What will be the output of the following recursive function?

```
int fun(int n) {
   if (n == 0) return 0;
   else return n + fun(n - 1);
}
System.out.println(fun(5));
```

//just write the answer

(iii) Which of the following is a reserved keyword in Java?
(a) string   (b) main   (c) final   (d) system

(iv) The propositional connective that results in true only if both operands are true is:
(a) Disjunction (∨)   (b) Negation (~)   (c) Implication (⇒)   (d) Conjunction (∧)

(v) The encoding scheme used by Java for multiple languages is:
(a) ASCII   (b) EBCDIC   (c) Unicode   (d) ISCII

(vi) Which loop executes at least once regardless of the condition?
(a) for   (b) while   (c) do-while   (d) nested for

(vii) A full adder circuit can be constructed using:
(a) Two AND + one OR   (b) Two XOR + one AND   (c) Two half adders + one OR   (d) Two half adders + one AND

(viii) Value of x after execution:
int x = 10; x += x++ + ++x;
(a) 32   (b) 31   (c) 22   (d) 21

(ix) Hexadecimal equivalent of $11010101_2$:
(a) $C5_{16}$   (b) $D5_{16}$   (c) $A5_{16}$   (d) $B5_{16}$

(x) Assertion (A): A constructor in Java must have the same name as the class.
Reason (R): The constructor initializes instance variables when an object is created.
(a) Both A and R true, R explains A
(b) Both A and R true, but R does not explain A
(c) A true, R false
(d) A false, R true

### Question 2 [5 × 2 = 10]

(a) Subtract 10110 from 11001 using the 2's complement method.
(b) Prove X + X′Y = X + Y using a truth table.
(c) Find the output of:
int m = 5, n = 2;

```
for (int i = 1; i <= m; i++) {
   if (i % n == 0) continue;
   System.out.print(i + " ");
}
```

(d) Write the WFF for: "It is neither cold nor raining." (P = "It is cold", Q = "It is raining").
(e) In recursion, what is the statement (or condition) that stops further recursive calls called? Why is it necessary?

## PART II (40 Marks)

Answer four questions from this part, choosing two from Section A and two from Section B.

## SECTION A (Answer any two) [20 Marks]

### Question 3

Vault security system design with four inputs (P, V, S, T).

Tasks: Truth table, SOP expression, K-map reduction, Logic circuit.

### Question 4

Simplify Boolean expression $F(A,B,C)=(A+B)(A'+C)(B+C)$ using Boolean laws. Draw Full Adder circuit and write expressions for Sum and Carry.

### Question 5

Prove NAND as a universal gate (construct NOT, AND, OR using NAND). Perform number conversions:

(i) $(1AF.C)_{16}$ to Binary
(ii) $(345)_8$ to Decimal
(iii) $(98.75)_{10}$ to Binary

## SECTION B (Answer any one) [10 Marks]

### Question 6

Java class ElectricityBill – Accept consumer details, compute bill using slabs, surcharge, and fixed charge, display final bill.

### Question 7

Java class PerfectNumber – Check if a number is perfect (sum of factors excluding itself equals the number).

## Additional Questions (Arrays, Loops, Recursion) – MUST ATTEND [10 Marks]

### Question 9

Java program to input an array of integers and print the second largest element.

### Question 10

Java program using nested loops to print a pyramid pattern of stars. Example (n=4):

```
   *
  ***
 *****
*******
```

### Question 11

Java recursive function to compute factorial of a number. Accept number, call recursive method, and display result.